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PAPER_875 – SM_mag Surface Conduction and Fragment Assembly

****CP4 Class:**** #459 `SMMagSurfaceConductionFragmentAssemblyCalc`

****Session:**** 200C v5.61

****Source:**** `describe mass without using weight.txt` (3,094 lines)

Abstract

SM_magnetic surface moments conduct from many surface points in a chaos pattern but spa
solid shell. SM_mag arranges brittle layered string fragments on the durable proto-
nucleus. Vacuum energy density becomes a fixed constant (capacitance) at proto-
nucleus formation. ULF quantum ripples $ULF_{quantum}^{-1..-26}$.

Key Equations

1. $C_{vac} = \rho_{vac} * r$ (vacuum energy density = capacitance)
2. $E_{crack} = \sum_{i=1}^{26} (\hbar * \omega / i) * C_{vac}$ (ULF quantum ripples)
3. $SM_{mag} = f_{SCm} * \rho_{SCm} * c^2$ (surface conduction strength)
4. $coherence = 1 - e^{-f_{SCm} * N_{frag}}$ (chaos -> coherent transition)
5. g_{atomic_SM} proportional to $1/(\rho_{vac} * r^2)$ (inversely proportional to vacuum density)

UQFF Context

This calculator implements parameterized physics equations per the
CondensedPhysics architecture (pure calculator, no hardcoded data).
Inputs arrive from source2.cpp PRINCIPAL GUI via dataset dict.
Outputs include primary_equations, available_equations, and simulation_set.

Source Thread

Grok 3 dialogue, Daniel Murphy, June 03-04 2025.
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Star-Magic UQFF Codebase – Session 200C

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.